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Inventor(s)	Polakowski; Matthew R et al.

Battery pack charger

Abstract

A temperature controlled enclosure that includes a temperature control device for controlling the temperature within an internal cavity of the temperature controlled enclosure. The temperature controlled enclosure also includes one or more charging ports for receiving and charging a battery pack. A controller within the temperature controlled enclosure controls the temperature within the internal cavity to a predetermined or desired temperature (e.g., 20° C.). When a battery pack is received in the one or more charging ports, the temperature of the battery pack can be determined. If, for example, the temperature of the battery pack is below 0° C., the battery pack is allowed to warm up inside the temperature controlled enclosure before the battery pack is charged.

Inventors: Polakowski; Matthew R (Greenfield, WI), Fassbender; Kyle C. (Brookfield, WI)

Applicant: MILWAUKEE ELECTRIC TOOL CORPORATION (Brookfield, WI)

Family ID: 1000008818541

Assignee: Milwaukee Electric Tool Corporation (Brookfield, WI)

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Primary Examiner: Erwin; James M

Attorney, Agent or Firm: Michael Best & Friedrich LLP

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 18/192,260, filed, Mar. 29, 2023, which is a continuation of U.S. patent application Ser. No. 17/962,742, filed Oct. 10, 2022, which is a continuation of U.S. patent application Ser. No. 17/673,391, filed Feb. 16, 2022, now U.S. Pat. No. 11,605,850, which is a continuation of U.S. patent application Ser. No. 16/776,623, filed on Jan. 30, 2020, now U.S. Pat. No. 11,283,117, which claims the benefit of U.S. Provisional Patent Application No. 62/798,583, filed on Jan. 30, 2019, the entire content of each of which is hereby incorporated by reference.

SUMMARY

(1) Embodiments described herein provide a temperature controlled enclosure that includes a temperature control device for controlling the temperature within an internal cavity of the temperature controlled enclosure. The temperature controlled enclosure also includes one or more charging ports for receiving and charging battery packs. A controller within the temperature controlled enclosure controls the temperature within the internal cavity to a predetermined or desired temperature (e.g., 20° C.). When a battery pack is received in the one or more charging ports, the temperature of the battery pack can be determined. If, for example, the temperature of the battery pack is below zero, the battery pack is allowed to warm up inside the temperature controlled enclosure before the battery pack is charged.

(2) Temperature controlled enclosures described herein include a port configured to receive a battery pack, a temperature control device configured to modify an ambient temperature within the enclosure, and an electronic controller. The electronic controller is configured to determine whether the battery pack is received in the port, receive a signal from the battery pack, the signal including at least one of a temperature signal and a charge enable signal, and control the ambient temperature within the enclosure using the temperature control device based on the signal received from the battery pack.

(3) Methods described herein for controlling a temperature within an enclosure include determining, using an electronic controller, when a battery pack is received in a port, receiving, using the electronic controller, a signal from the battery pack, the signal including at least one of a temperature signal and a charge enable signal, and controlling, using a temperature control device, the temperature within the enclosure based on the signal.

(4) Temperature controlled enclosures described herein include a plurality of ports with each port of the plurality of ports being configured to receive a battery pack, a control device configured to

modify an ambient temperature within the enclosure, and an electronic controller. The electronic controller is configured to determine whether the battery pack is received in a port of the plurality of ports, receive a signal from the battery pack, the signal including at least one of a temperature signal and a charge enable signal, and control the ambient temperature within the enclosure using the temperature control device based on the signal received from the battery pack.

(5) Before any embodiments are explained in detail, it is to be understood that the embodiments are not limited in its application to the details of the configuration and arrangement of components set forth in the following description or illustrated in the accompanying drawings. The embodiments are capable of being practiced or of being carried out in various ways. Also, it is to be understood that the phraseology and terminology used herein are for the purpose of description and should not be regarded as limiting. The use of “including,” “comprising,” or “having” and variations thereof are meant to encompass the items listed thereafter and equivalents thereof as well as additional items. Unless specified or limited otherwise, the terms “mounted,” “connected,” “supported,” and “coupled” and variations thereof are used broadly and encompass both direct and indirect mountings, connections, supports, and couplings.

(6) In addition, it should be understood that embodiments may include hardware, software, and electronic components or modules that, for purposes of discussion, may be illustrated and described as if the majority of the components were implemented solely in hardware. However, one of ordinary skill in the art, and based on a reading of this detailed description, would recognize that, in at least one embodiment, the electronic-based aspects may be implemented in software (e.g., stored on non-transitory computer-readable medium) executable by one or more processing units, such as a microprocessor and/or application specific integrated circuits (“ASICs”). As such, it should be noted that a plurality of hardware and software based devices, as well as a plurality of different structural components, may be utilized to implement the embodiments. For example, “servers” and “computing devices” described in the specification can include one or more processing units, one or more computer-readable medium modules, one or more input/output interfaces, and various connections (e.g., a system bus) connecting the components.

(7) Other aspects of the embodiments will become apparent by consideration of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view of a temperature controlled enclosure, according to embodiments described herein.

(2) FIG. 2 is a front view of the temperature controlled enclosure of FIG. 1.

(3) FIG. 3 is a rear view of the temperature controlled enclosure of FIG. 1.

(4) FIG. 4 is a top view of the temperature controlled enclosure of FIG. 1.

(5) FIG. 5 is a bottom view of the temperature controlled enclosure of FIG. 1.

(6) FIG. 6 is a left-side view of the temperature controlled enclosure of FIG. 1.

(7) FIG. 7 is a right-side view of the temperature controlled enclosure of FIG. 1.

(8) FIG. 8 is a perspective view of a device including an interface for mechanically engaging with the temperature controlled enclosure of FIG. 1, according to embodiments described herein.

(9) FIG. 9 is a top view of the temperature controlled enclosure of FIG. 1 with an upper housing portion removed, according to embodiments described herein.

(10) FIG. 10 is an electromechanical schematic diagram of a controller for the temperature controlled enclosure of FIG. 1, according to embodiments described herein.

(11) FIG. 11 illustrates a temperature control device for the temperature controlled enclosure of FIG. 1, according to embodiments described herein.

(12) FIG. 12 is a partial cross-sectional view of a wall of a housing of the temperature controlled enclosure of FIG. 1, according to embodiments described herein.

(13) FIG. 13 is a process for controlling the temperature controlled enclosure of FIG. 1, according to embodiments described herein.

DETAILED DESCRIPTION

(14) FIGS. 1-7 illustrate a temperature controlled enclosure **100** that includes a lower housing portion **105** and an upper housing portion **110**. In some embodiments, the upper housing portion **110** is pivotable about one or more hinges such that the interior of the temperature controlled enclosure **100** can be accessed without completely removing the upper housing portion **110**. The upper housing portion **110** is configured to engage the lower housing portion **105** to create a substantially air- and water-tight seal. Such a seal between the lower housing portion **105** and the upper housing portion **110** aides in preventing air inside the temperature controlled enclosure **100** from being affected by air outside of the temperature controlled enclosure **100**.

(15) The temperature controlled enclosure **100** also includes a power input terminal **115**. In some embodiments, the power input terminal **115** is an AC power input terminal. In other embodiments, the power input terminal **115** is a DC power input terminal or includes a battery pack interface for receiving one or more battery packs. The upper housing portion **110** includes an interface **120** that is configured to allow the temperature controlled enclosure **100** to physically engage or mate with one or more additional devices that have a complimentary interfaces.

(16) With reference to FIG. 4, the interface **120** of the upper housing portion **110** includes a plurality of connection recesses **125** that receive and cooperate with projections from a complementary interface (see FIG. 8). The connection recesses **125** include two rows of two small recesses **130** and one row of large recess **135**. In other embodiments, the upper housing portion **110** can include different numbers or patterns of connection recesses **125**. An interference projection or wing **140** extends into each connection recess **125** on opposite sides of the connection recess **125** from one end of the connection recess **125**. Each of the wings **140** has a length that extends approximately half the connection recess **125** to define a first portion of the connection recess **125** and a second portion opposite the wings **140**, which remains open. In some embodiments, each of the wings **140** has a length that extends less than half the connection recess **125**.

(17) With reference to a device **200** (e.g., a component of a modular tool storage system) having a bottom member **205** as illustrated in FIG. 8, the second portion of each connection recess **125** is sized to receive a projection **210**. When the projections **210** are received in the connection recesses **125**, a bottom surface **215** of the bottom member **205** is arranged to contact and be supported by a top surface of the upper housing portion **110**. In the illustrated embodiment, each projection **210** extends from the bottom surface **215** of the bottom member **205** and is configured to cooperate with a connection recess **125** and connect the device **200** to the temperature controlled enclosure **100**. In the illustrated embodiment, each projection **210** is formed integrally with the bottom member **205**. Each projection **210** has a channel **220** on each side of the projection **210** extending parallel to a longitudinal axis A of the bottom member **205**. Each channel **220** has a front, open end **225** and a back, closed end **230** along the axis A. Each projection **210** has a planar surface **235** with a generally rectangular shape. In other embodiments, the planar surface **235** may be another shape (e.g., circular, triangular, etc.). In the illustrated embodiment, there are six projections **210** arranged in three rows of two along the axis A. In other embodiments, the device **200** includes more or fewer projections **210**, and/or the projections **210** are arranged in different patterns. The projections **210** are arranged such that the device **200** can be supported on a surface by the projections **210** through contact with the planar surfaces **235**. In some embodiments, the device **200** includes an interface portion **240** and a latch member **245** for engaging a locking aperture of the temperature controlled enclosure **100** and securing the device **200** to the temperature controlled enclosure **100**.

(18) Each of the wings **140** of the connection recesses **125** corresponds to and is configured to cooperate with a corresponding one of the channels **220** of the projection **210** received by the

respective connection recess **125**. In the disconnected position, the projections **210** are oriented within the connection recesses **125** such that the open ends **225** of the channels **220** are nearer to the wings **140** than the closed ends **230**. Once in the disconnected position, the bottom member **205** may slide relative to the upper housing portion **110** parallel the longitudinal axis A in a first direction toward the wings **140** such that the wings **140** are received within the channels **220** in a second, interfaced or connected position. The wings **140** and the projections **210** engage within the connection recesses **125** to interface and connect the bottom member **205** with the upper housing portion **110** and prevent disconnection of the bottom member **205** from the upper housing portion **110**, except in a second direction opposite the first direction and generally parallel to the longitudinal axis A. The wings **140** and the channels **220** engage one another perpendicular to the longitudinal axis A. In some embodiments, the upper housing portion **110** includes a locking aperture for inhibiting disconnection of the device **200** from the temperature controlled enclosure **100**.

(19) FIG. **9** illustrates the temperature controlled enclosure **100** with the upper housing portion **110** removed. As shown in FIG. **9**, the temperature controlled enclosure **100** includes an internal cavity or space **145**. The internal cavity **145** includes one or more battery pack interfaces **150** (e.g., battery pack charging ports). The battery pack charging ports **150** correspond to charging ports for a first type of battery pack (e.g., a stem battery pack) including a first output voltage (e.g., 12V). The internal cavity **145** also includes one or more battery pack charging ports **155**. The battery pack charging ports **155** correspond to charging ports for a second type of battery pack (e.g., a brick or slide-on battery pack) including a second output voltage (e.g., 18V). The internal cavity **145** also includes one or more indicators for providing a charging and/or temperature status of one or more battery packs connected to the charging ports **150**, **155**.

(20) A controller **300** for the temperature controlled enclosure **100** is illustrated in FIG. **10**. The controller **300** is electrically and/or communicatively connected to a variety of modules or components of the temperature controlled enclosure **100**. For example, the illustrated controller **300** is connected the indicators **160**, the battery pack charging ports **150** and the battery pack charging ports **155** through a first power control module **305** and a second power control module **310**, respectively, a temperature control device **315** (e.g., a forced air temperature control device), one or more temperature sensors **320** (e.g., thermistors for sensing ambient temperature, a temperature of a battery pack, a temperature of a charging port, etc.), and a power input circuit **325**. The controller **300** includes combinations of hardware and software that are operable to, among other things, control the operation of the temperature controlled enclosure **100**, activate the indicators **160** (e.g., one or more LEDs), receive signals from the one or more temperature sensors **320**, control the temperature control device **315**, control the power control modules **305**, **310**, etc. In some embodiments, the indicators **160** also include one or more indicators that are visible externally to the temperature controlled enclosure **100** (e.g., positioned on an external surface of the lower housing portion). Such indicators can provide an indication of an internal ambient temperature of the temperature controlled enclosure **100** or a battery pack charging status.

(21) The controller **300** includes a plurality of electrical and electronic components that provide power, operational control, and protection to the components and modules within the controller **300** and/or temperature controlled enclosure **100**. For example, the controller **300** includes, among other things, a processing unit **330** (e.g., a microprocessor, electronic controller, electronic processor, a microcontroller, or another suitable programmable device), a memory **335**, input units **340**, and output units **345**. The processing unit **330** includes, among other things, a control unit **350**, an arithmetic logic unit (“ALU”) **325**, and a plurality of registers **360** (shown as a group of registers in FIG. **10**), and is implemented using a known computer architecture (e.g., a modified Harvard architecture, a von Neumann architecture, etc.). The processing unit **330**, the memory **335**, the input units **340**, and the output units **345**, as well as the various modules connected to the controller **300** are connected by one or more control and/or data buses (e.g., common bus **365**). The

control and/or data buses are shown generally in FIG. 10 for illustrative purposes. The use of one or more control and/or data buses for the interconnection between and communication among the various modules and components would be known to a person skilled in the art in view of the invention described herein.

(22) The memory **335** is a non-transitory computer readable medium and includes, for example, a program storage area and a data storage area. The program storage area and the data storage area can include combinations of different types of memory, such as a ROM, a RAM (e.g., DRAM, SDRAM, etc.), EEPROM, flash memory, a hard disk, an SD card, or other suitable magnetic, optical, physical, or electronic memory devices. The processing unit **330** is connected to the memory **335** and executes software instructions that are capable of being stored in a RAM of the memory **335** (e.g., during execution), a ROM of the memory **335** (e.g., on a generally permanent basis), or another non-transitory computer readable medium such as another memory or a disc. Software included in the implementation of the temperature controlled enclosure **100** can be stored in the memory **335** of the controller **300**. The software includes, for example, firmware, one or more applications, program data, filters, rules, one or more program modules, and other executable instructions. The controller **300** is configured to retrieve from the memory **335** and execute, among other things, instructions related to the control processes and methods described herein. In other constructions, the controller **300** includes additional, fewer, or different components.

(23) As shown in FIG. 9, the battery pack charging ports **150**, **155** include a combination of mechanical components and electrical components configured to and operable for interfacing (e.g., mechanically, electrically, and communicatively connecting) a battery pack with the temperature controlled enclosure **100**. For example, the battery pack charging ports **150**, **155** are configured to receive power from the power control modules **305**, **310**, respectively, via power lines between the power control modules **305**, **310** and the battery pack charging ports **150**, **155**. The battery pack charging ports **150**, **155** are also configured to communicatively connect to the power control modules **305**, **310** via separate communications lines.

(24) The one or more temperature sensors **320** include one or more thermistors or other temperature sensors that are operable to monitor a temperature within the temperature controlled enclosure **100** (e.g., an ambient temperature in the internal cavity **145**), a temperature of one or more battery packs, a temperature of one or more charging ports **150**, **155**, etc. The one or more temperature sensors **320** provide output signals (e.g., temperature output signals) to the controller **300**. Based on these control signals, the controller **300** determines the ambient air temperature within the temperature controlled enclosure **100**, the temperature of one or more battery packs, the temperature of one or more charging ports **150**, **155**, etc. The controller **300** is configured to control the operation of the temperature control device **315** to regulate the temperature of the internal cavity **145** of the temperature controlled enclosure **100** and charge one or more battery packs.

(25) The power input circuit **325** includes the power input terminal **115**. As previously described, the temperature controlled enclosure **100** can be powered by AC power (e.g., AC mains power) or DC power (e.g., a battery pack). In some embodiments, a DC power source (e.g., a battery pack) powers an inverter which then provides AC power to the power input terminal **115**. In some embodiments, the temperature controlled enclosure **100** is configured to operate using approximately 150 Watts of power. For example, the temperature controlled enclosure **100** using 150 Watts of power is able to maintain a predetermined or desired ambient temperature in the internal cavity **145** of the temperature controlled enclosure **100** when an ambient air temperature external to the temperature controlled enclosure **100** is approximately minus 20° C. In some embodiments, the predetermined or desirable ambient temperature in the internal cavity **145** of the temperature controlled enclosure **100** is between approximately 20° C. and approximately 30° C. In such a configuration, the temperature controlled enclosure **100** is operable to raise the temperature of a battery pack that is at a temperature of minus 20° C. to 0° C. in approximately ninety minutes. In some embodiments, the temperature controlled enclosure **100** also includes a burst mode in

which an ambient temperature of the internal cavity **145** is increased to approximately 50° C. and the temperature of any battery packs within the temperature controlled enclosure **100** is increased more quickly. In such embodiments, the temperature controlled enclosure **100** is configured to operate using approximately 265 Watts of power.

(26) FIG. **11**. illustrates an internal portion of the temperature controlled enclosure **100**, according to embodiments described herein. The internal portion of the temperature controlled enclosure **100** includes, among other things, the controller **300** and the temperature control device **315**. In the illustrated embodiment, the temperature control device **315** is a thermoelectric cooling device, such as a Peltier device. The temperature control device **315** is configured to regulate the temperature of the temperature controlled enclosure **100**. The temperature control device **315** is capable of either generating heat (i.e., to increase or maintain a temperature) or absorb heat (i.e., to decrease or maintain a temperature). In some embodiments, the temperature control device **315** can also include a fan to assist in circulating heated or cooled air. In some embodiments, the temperature control device **315** includes a resistive heating element and a fan or another forced air heating system. In such embodiments, heat generated by the resistive heating element is circulated within the temperature controlled enclosure **100** using the fan.

(27) To increase the efficiency of the temperature controlled enclosure **100**, the temperature controlled enclosure **100** can also be insulated. FIG. **12** illustrates a partial cross-section **400** of part of the lower housing portion **105** of the temperature controlled enclosure **100**. The partial cross-section **400** includes three layers of materials. A first insulating layer of material **405** corresponds to an outer housing material and is made of a plastic, such as polypropylene, high-impact polystyrene, or another similar material. The first layer of material **405** has a thermal resistance (“R”) value of approximately 0.025. A second insulating layer of material **410** corresponds to an inner material layer for the internal cavity **145** of the temperature controlled enclosure **100**. The second layer of material **410** is made of reflective insulation and has an R-value of between approximately 1.0 and 1.5. In some embodiments, a third insulating layer of material **415** is placed between the first layer of material **405** and the second layer of material **410**. The third layer of material **415** is another insulating material that further prevents conduction of heat through the temperature controlled enclosure **100** and increase an overall R-value of the temperature controlled enclosure **100**. In some embodiments, the R-value of the temperature controlled enclosure **100** is approximately 1.25-1.3. In other embodiments, the R-value for the temperature controlled enclosure **100** is in the range of 1.0-2.0.

(28) FIG. **13** is a process **500** for controlling the temperature controlled enclosure **100**, according to embodiments described herein. The steps of the process **500** are shown for illustrative purposes. The process **500** can perform one of more of the steps in an order different than that shown in FIG. **13**, or one or more steps of the process **500** can be removed from the process **500**. The process **500** begins with the temperature controlled enclosure **100** receiving power from a power source (e.g., through the power input terminal **115**) (STEP **505**). At step **510**, the controller **300** determines whether temperature control is ON or enabled. In some embodiments, a user manually turns temperature control ON (e.g., using a temperature control switch). In other embodiments, temperature control is automatically turned ON when the temperature controlled enclosure **100** receives power. If temperature control is OFF, the process waits for temperature control to be turned ON at STEP **510**. If, at STEP **510**, temperature control is ON, the controller **300** determines whether a battery pack is connected to a charging port within the temperature controlled enclosure **100** (STEP **515**). If no battery pack is connected to a charging port, the process **500** returns to STEP **510**. If a battery pack is connected to a charging port, the controller **300** detects the temperature of the battery pack (STEP **520**). The controller **300** determines the temperature of the battery pack using a temperature sensor (e.g., a thermistor) in proximity to battery pack. In some embodiments, if the battery pack is capable of communication, the battery pack is configured to communicate the battery pack temperature (e.g., a temperature signal) to the controller **300**. In some embodiments,

the battery pack transmits a charge enable signal to the controller **300**. The charge enable signal is configured to indicate the battery pack is, for example, within an acceptable temperature range for charging. In some embodiments, the process **500** does not include determining whether a battery pack is connected to a charging port at STEP **515** or determine a battery pack temperature at STEP **520** (i.e., the temperature controlled enclosure **100** independently regulates the ambient temperature of the internal cavity **145**).

(29) After STEP **520**, the controller **300** operates the temperature control device **315** to control the ambient temperature in the internal cavity **145** of the temperature controlled enclosure **100** (STEP **525**). When the temperature control device **315** is operated, the controller **300** can set a timer such that the temperature control device **315** is operated for a predetermined or desired amount of time (STEP **530**). After the timer is set at STEP **530**, the controller **300** determines if the temperature of the battery pack is greater than or equal to a temperature threshold value (e.g., 0° C.) (STEP **535**). If the battery pack temperature is not greater than or equal to the temperature threshold value, the controller **300** compares the timer to a timer threshold value (e.g., 90 minutes) (STEP **540**). If the timer is not greater than or equal to the timer threshold value, the process **500** returns to step **530**. If, at STEP **535**, the battery pack temperature is greater than or equal to the temperature threshold value, or, at STEP **540**, the timer is greater than or equal to the timer threshold value, the process **500** proceeds to STEP **545** where the temperature control device **315** is disabled, and the battery pack is charged (STEP **550**). In some embodiments, the temperature control device **315** is not disable at STEP **545** and continues to be operated when a battery pack is being charged.

(30) Thus, embodiments described herein provide, among other things, a temperature controlled enclosure for charging one or more battery packs.

Claims

1. A charger comprising: an interface configured to receive a battery pack; a temperature control device configured to increase a temperature of the battery pack; and an electronic controller configured to: determine whether the battery pack is received by the interface, receive a signal from the battery pack, the signal including at least one of a temperature signal and a charge enable signal, determine a temperature of the battery pack based on the signal, control the temperature of the battery pack using the temperature control device based on the signal received from the battery pack, set a timer indicating a period of time during which the temperature control device is activated, compare the timer to a timer threshold value, disable the temperature control device when the timer is greater than or equal to the timer threshold value, compare the temperature of the battery pack to a temperature threshold value, disable the temperature control device when the temperature of the battery pack is greater than or equal to the temperature threshold value, and charge the battery pack.
2. The charger of claim 1, wherein the electronic controller is further configured to: determine whether a temperature control switch is set to ON; and operate, when both the temperature control switch is set to ON and the battery pack is received by the interface, the temperature control device.
3. The charger of claim 1, further comprising: a power input terminal configured to receive power from an alternating current (AC) power source.
4. The charger of claim 1, wherein the temperature control device includes a fan.
5. The charger of claim 1, further comprising: an indicator configured to provide an indication of a charging status of the battery pack.
6. The charger of claim 1, wherein the temperature control device includes a resistive heating element.
7. The charger of claim 1, further comprising: a housing including: a first insulating layer corresponding to an outer material layer; a second insulating layer corresponding to an inner material layer; and a third insulating layer positioned between the first insulating layer and the

second insulating layer.

8. The charger of claim 7, wherein the housing includes a thermal resistance value between 1.0 and 2.0.

9. The charger of claim 1, the charger further comprising: a lower housing portion; and an upper housing portion configured to engage the lower housing portion, wherein the upper housing portion and the lower housing portion create an air-tight seal.

10. A method of controlling a temperature of a battery pack, the method comprising: determining, using an electronic controller, when a battery pack is received in an interface; receiving, using the electronic controller, a signal from the battery pack, the signal including at least one of a temperature signal and a charge enable signal; determining, using the electronic controller, a temperature of the battery pack based on the signal; controlling, using a temperature control device, the temperature of the battery pack based on the signal; setting a timer indicating a period of time during which the temperature control device is activated; comparing the timer to a timer threshold value; disabling the temperature control device when the timer is greater than or equal to the timer threshold value; comparing the temperature of the battery pack to a temperature threshold value; disabling the temperature control device when the temperature of the battery pack is greater than or equal to the temperature threshold value; and charging the battery pack.

11. The method of claim 10, further comprising: determining whether a temperature control switch is set to ON, and operating, when both the temperature control switch is set to ON and the battery pack is received by the interface, the temperature control device.

12. A charger comprising: a plurality of interfaces, each interface of the plurality of interfaces configured to receive a battery pack; a temperature control device configured to modify a temperature of the battery pack; and an electronic controller configured to: determine whether the battery pack is received by an interface of the plurality of interfaces, receive a signal from the battery pack, the signal including at least one of a temperature signal and a charge enable signal, determine a temperature of the battery pack based on the signal, control the temperature of the battery pack using the temperature control device based on the signal received from the battery pack, set a timer indicating a period of time during which the temperature control device is activated, compare the timer to a timer threshold value, disable the temperature control device when the timer is greater than or equal to the timer threshold value, compare the temperature of the battery pack to a temperature threshold value, disable the temperature control device when the temperature of the battery pack is greater than or equal to the temperature threshold value, and charge the battery pack.

13. The charger of claim 12, wherein a first interface of the plurality of interfaces is configured to receive a first battery pack type, the first battery pack type including a first output voltage.

14. The charger of claim 13, wherein a second interface of the plurality of interfaces is configured to receive a second battery pack type, the second battery pack type including a second output voltage, the second output voltage different from the first output voltage.

15. The charger of claim 12, further comprising: an indicator associated with a first interface of the plurality of interfaces, the indicator configured to indicate a charging status of the battery pack received by the first interface.

16. The charger of claim 12, further comprising: a power input terminal configured to receive power from an alternating current (AC) power source.

17. The charger of claim 12, wherein the temperature control device includes a fan.

18. The charger of claim 12, wherein the temperature control device includes a resistive heating element.
